

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4,BL5)
Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks:35

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _____Delisha A K (192511307)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Delisha

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving

Q1. Scenario: Online Banking System

System Components:

Account Management, Fund Transfer, Bill Payment, Transaction History

Given Constraints:

- Transaction completion time < 2 seconds
- Multi-factor authentication required
- Support 5000+ concurrent users

Question:

Analyze the system using quality assurance techniques under the given constraints. Identify potential defects related to security, performance, and reliability. Propose suitable testing strategies and justify how they ensure system quality. **(BL4)**

Online Banking System – Quality Assurance Analysis

Problem Understanding

The Online Banking System includes **Account Management, Fund Transfer, Bill Payment, and Transaction History**. The system must complete transactions within **2 seconds**, use **multi-factor authentication**, and support **5000+ concurrent users**.

Constraint Analysis

- **Security:** MFA must prevent unauthorized access.
- **Performance:** Transactions must finish within 2 seconds.
- **Scalability:** The system must handle 5000+ users.
- **Reliability:** Transactions and account balances must remain accurate.

Potential Defects

- Unauthorized account access.
- Weak MFA or authentication.
- Slow transaction response.
- Server failure under heavy load.
- Duplicate or incomplete transactions.
- Incorrect account balance or transaction history.

Testing Strategies

- **Security Testing:** Check MFA, login, authorization, and data protection.
- **Functional Testing:** Verify fund transfer, bill payment, and transaction history.
- **Performance Testing:** Ensure transactions complete within 2 seconds.
- **Load Testing:** Test the system with 5000+ simultaneous users.
- **Reliability Testing:** Check for duplicate, failed, or incorrect transactions.

Justification

These testing techniques directly address the given constraints. Security testing protects customer accounts, performance testing ensures fast transactions, load testing verifies support for 5000+ users, and reliability testing ensures accurate financial transactions.

Conclusion

A combination of **security, functional, performance, load, and reliability testing** will help make the online banking system **secure, fast, scalable, and reliable**.

Q2. Scenario: E-Commerce Shopping Application

System Features:

Product Search, Shopping Cart, Checkout, Payment Gateway

Usability Constraints:

- Checkout process within 3 steps
- Easy product navigation

- Responsive design for mobile and desktop devices

Question:

Apply usability testing techniques to evaluate the application. Identify usability challenges and recommend design improvements that satisfy the given constraints. Explain how the proposed changes improve customer satisfaction and efficiency. **(BL4)**

E-Commerce Shopping Application – Usability Testing

Problem Understanding

The application includes **Product Search, Shopping Cart, Checkout, and Payment Gateway**. The checkout should be completed within **3 steps**, navigation should be easy, and the interface must work well on **mobile and desktop devices**.

Usability Challenges

- Users may find products difficult to locate.
- Too many checkout steps may confuse users.
- Navigation may be unclear.
- The interface may not display properly on mobile devices.
- Complicated payment forms may increase errors.

Usability Testing Techniques

- **User Testing:** Observe real users while searching, adding products, and completing purchases.
- **Task-Based Testing:** Ask users to complete tasks such as finding a product and placing an order.
- **A/B Testing:** Compare different layouts to identify the easier design.
- **Responsive Testing:** Check the application on different mobile and desktop screen sizes.

Design Improvements

- Use a clear search bar and simple product categories.

- Keep checkout limited to **three simple steps**.
- Provide clear buttons such as **Add to Cart** and **Place Order**.
- Use a responsive design for mobile and desktop.
- Display clear error messages and payment confirmation.

Justification

These improvements reduce user confusion, make navigation easier, and allow customers to complete purchases faster. A simple and responsive interface improves **customer satisfaction, usability, and shopping efficiency**.

Conclusion

Usability testing helps identify difficulties faced by customers. By providing **simple navigation, a three-step checkout, responsive design, and clear feedback**, the application becomes easier, faster, and more convenient to use.

Q3. Scenario: Hospital Management System

Modules:

Patient Registration, Appointment Scheduling, Medical Records, Billing

QA Constraints:

- Patient data accuracy must be maintained
- System availability 24/7
- Simultaneous access by multiple departments

Question:

Use quality assurance practices to analyze the system. Identify critical test cases and discuss how the constraints influence testing priorities, defect prevention, and overall system reliability.

(BL5)

Hospital Management System – Quality Assurance Analysis

Problem Understanding

The Hospital Management System includes **Patient Registration, Appointment Scheduling, Medical Records, and Billing**. The system must maintain accurate patient data, provide **24/7 availability**, and allow multiple departments to access the system simultaneously.

Critical Test Cases

- Verify that patient details are entered and stored accurately.
- Check appointment booking, cancellation, and rescheduling.
- Verify that medical records are correctly stored and retrieved.
- Check billing calculations and payment records.
- Test simultaneous access by different departments.
- Check system recovery after failures.

Quality Assurance Practices

- **Functional Testing:** Verify that all hospital modules work correctly.
- **Data Validation Testing:** Ensure patient information and medical records remain accurate.
- **Concurrency Testing:** Check whether multiple departments can access records without conflicts.
- **Availability Testing:** Ensure the system remains operational 24/7.
- **Recovery Testing:** Verify that data is not lost after system or network failures.

Impact of Constraints

Data accuracy has the highest priority because incorrect patient information can affect healthcare decisions. **24/7 availability** requires continuous monitoring and recovery mechanisms. **Simultaneous access** requires concurrency testing to prevent data conflicts and ensure consistent records.

Conclusion

Quality assurance should focus on **data accuracy, continuous availability, and safe concurrent access**. These testing practices help prevent critical defects and ensure a reliable hospital management system

Q4. Scenario: Smart Library Management System

Features:

Book Search, Book Reservation, Borrow/Return, Notifications

Usability Constraints:

- Search results displayed within 2 seconds
- Accessible interface for all age groups
- Minimal learning curve for first-time users

Question:

Conduct a usability evaluation using appropriate testing methods. Analyze how the given constraints impact interface design and recommend improvements to enhance usability and accessibility. **(BL5)**

Smart Library Management System – Usability Evaluation

Problem Understanding

The Smart Library Management System includes **Book Search, Book Reservation, Borrow/Return, and Notifications**. Search results should appear within **2 seconds**, the interface should be accessible to all age groups, and first-time users should require minimal learning.

Usability Challenges

- Search results may be slow.
- Older users may find the interface difficult to understand.
- Too many options can confuse first-time users.

- Small text or unclear buttons may reduce accessibility.

Usability Testing Methods

- **User Testing:** Observe users while searching and reserving books.
- **Task-Based Testing:** Ask users to perform common tasks such as borrowing and returning books.
- **Accessibility Testing:** Check text size, button visibility, readability, and ease of navigation.
- **Performance Testing:** Ensure search results appear within 2 seconds.

Recommended Improvements

- Provide a simple search bar with clear categories.
- Use large, readable text and clearly visible buttons.
- Keep navigation simple and consistent.
- Provide clear instructions and feedback messages.
- Make search and reservation processes quick and easy.

Justification

These improvements reduce the learning curve and make the system easier for users of different age groups. Fast search results improve efficiency, while accessible design improves overall usability.

Conclusion

Usability testing helps identify difficulties faced by library users. A **simple, accessible, fast, and user-friendly interface** will improve both usability and user satisfaction.

Q5. Scenario: Airline Reservation System

Modules:

User Login, Flight Search, Seat Selection, Ticket Booking

QA & Usability Constraints:

- No booking duplication
- System must support peak holiday traffic
- Booking interface should be simple and error-free

Question:

Apply both quality assurance and usability testing techniques under the specified constraints. Identify major risks, propose suitable testing methods, and explain how the system can be optimized for reliability, performance, and user experience. **(BL4)**

Airline Reservation System – Quality Assurance and Usability Testing

Problem Understanding

The Airline Reservation System includes **User Login, Flight Search, Seat Selection, and Ticket Booking**. The system must prevent booking duplication, support peak holiday traffic, and provide a simple, error-free booking interface.

Major Risks

- Duplicate booking of the same seat.
- System slowdown during peak traffic.
- Incorrect flight or seat information.
- Errors during ticket payment or booking.
- Confusing booking interface.

Testing Methods

- **Functional Testing:** Verify login, flight search, seat selection, and ticket booking.
- **Concurrency Testing:** Ensure two users cannot book the same seat simultaneously.
- **Load Testing:** Test the system with heavy holiday traffic.
- **Usability Testing:** Check whether users can easily search, select seats, and book tickets.
- **Performance Testing:** Verify fast response during high traffic.

Recommended Improvements

- Use real-time seat availability updates.

- Prevent duplicate bookings using proper transaction validation.
- Increase server capacity for peak traffic.
- Provide a simple step-by-step booking interface.
- Display clear error and confirmation messages.

Justification

These techniques improve **reliability, performance, and usability**. Concurrency testing prevents duplicate bookings, load testing ensures the system handles peak traffic, and usability testing makes the booking process simple and error-free.

Conclusion

A combination of **quality assurance and usability testing** ensures that the airline reservation system is **reliable, fast, secure, and easy to use**, while satisfying all the given constraints.